Aperture Threshold Geometry in the SHA-256 Message Schedule

A Recurrence-Theoretic Classification of Seed-Word Visibility, Clean Echo Sets, Saturation Boundaries, and Projection States

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May 2026

Abstract

This paper develops a formal recurrence-theoretic model of the SHA-256 message schedule, called Aperture Threshold Geometry. The model studies the 16-word seed window $W[0..15]$ and the 48-word expansion zone $W[16..63]$ not as an unstructured diffusion field, but as a deterministic recurrence machine with exact visibility thresholds, clean echo sets, and downstream saturation boundaries. The core observation is that the SHA-256 schedule recurrence

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}$$

induces a four-point aperture complement

$$A = 16 - \{2,7,15,16\} = \{14,9,1,0\},$$

or, in increasing seed order,

$$A = \{0,1,9,14\}.$$

These coordinates are the structural phase-change boundaries of the seed window. They determine the direct-entry classes of seed words, the first expanded aperture readout at $W[16]$, and the later contamination geometry of the schedule.

The paper proves the Direct-Entry Spectrum Theorem, the Seed Echo Taxonomy Theorem, and the Seed Contamination Saturation Theorem. Together these results classify every seed coordinate $j \in \{0,\ldots,15\}$ by its direct-entry set $D(j)$, its clean echo set $\mathcal C_j$, and its saturation threshold S_j. The exact taxonomy is:

$$\begin{aligned} &j=0: &&\mathcal C_0=\{17,19,21\}, &&S_0=22,\\ &1\le j\le 8: &&\mathcal C_j=\{16,\ldots,j+14\}, &&S_j=j+15,\\ &9\le j\le 13: &&\mathcal C_j=\{16,\ldots,j+12\}\setminus\{j+7,j+9,j+11\}, &&S_j=j+13,\\ &j=14: &&\mathcal C_{14}=\{17,19\}, &&S_{14}=20,\\ &j=15: &&\mathcal C_{15}=\{16,18,20\}, &&S_{15}=21. \end{aligned}$$

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The coordinate $W[9]$ is then isolated as a distinguished hinge coordinate: it is the first seed word with triple direct-entry into the expansion lattice, with

$$D(9)=\{16,24,25\},$$

and it is also the length-sensitive vertex of the final-block seed triangle $T_C=\{0,1,9\}$. The $W[9]$ hinge enters the expansion as raw data at $W[16]$, as $\sigma_0(W[9])$ at $W[24]$, and as raw data again at $W[25]$. The interval $W[16]\to W[25]$ is a 9-step hinge envelope whose 8-word interior is the Sziklai window $W[16..23]$.

Finally, the paper defines a final-block projection layer through the fold-state vector $\Theta_9(r)$, where r is the number of message bytes carried into the final block. This vector separates three schedule faces: zero face, padding crease, and live content face. The resulting formulation reframes SHA-256 message expansion as three nested machines: an aperture machine, an echo/saturation machine, and a projection/shadow machine. A final section states a testable transport hypothesis relating the recurrence geometry to transformer attention mechanics, while explicitly distinguishing proven recurrence facts from cross-domain conjecture.

Keywords: SHA-256, message schedule, recurrence geometry, direct-entry spectrum, clean echo set, saturation threshold, aperture complement, W9 hinge law, seed echo taxonomy, cryptographic trace projection, transformer attention, standing wave geometry.

1. Introduction

SHA-256 is usually analyzed at the level of cryptographic security: collision resistance, preimage resistance, diffusion, and avalanche behavior. From that perspective, the internal message schedule is treated primarily as a mechanism for spreading input differences rapidly into later compression rounds. This abstraction is appropriate for standard security analysis, but it hides a separate mathematical object: the deterministic recurrence geometry of the schedule itself.

The SHA-256 message schedule begins with sixteen 32-bit seed words,

$$W[0],W[1],\ldots,W[15],$$

then expands them to sixty-four words by the recurrence

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}},\qquad 16\le t\le 63.$$

This recurrence has four lags:

$$L=\{2,7,15,16\}.$$

The central thesis of this paper is that the lag set does not merely specify an implementation recipe. It carves the seed window into a finite aperture geometry. When the lag set is reflected against the 16-word seed width, it yields the aperture complement

$$A = 16-L = \{14,9,1,0\}.$$

Sorted in seed order:

$$A=\{0,1,9,14\}.$$

These four positions are exactly the phase-change boundaries at which new recurrence slots first become available to seed words. The result is a complete entry taxonomy:

$$0\ \vert\ 1..8\ \vert\ 9..13\ \vert\ 14..15.$$

That partition is not heuristic. It follows directly from the inequalities required for a seed word to land in the expansion zone. The expansion zone begins at $t=16$; therefore, a seed word $W[j]$ can enter the expansion directly through a lag $\ell\in L$ only if

$$j+\ell \ge 16.$$

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The four threshold values $16-\ell$ are precisely $0,1,9,14$.

This paper develops the mathematical consequences of that observation. The results are exact recurrence statements. They do not claim a practical preimage attack against SHA-256. They also do not claim that the digest ceases to behave cryptographically. The target is narrower: to classify the schedule's internal visibility structure before ordinary avalanche saturation obscures the route by which each seed coordinate enters the expansion lattice.

1.1 Contributions

This paper makes five main contributions.

First, it defines the Aperture Complement Law for recurrence schedules and specializes it to SHA-256:

$$A=N-L,$$

where $N=16$ is the seed width and $L=\{2,7,15,16\}$ is the SHA-256 schedule lag set. This yields

$$A=\{0,1,9,14\}.$$

Second, it proves the Direct-Entry Spectrum Theorem, which classifies all seed words by direct-entry cardinality:

$$|D(j)|= \begin{cases} 1,&j=0,\\ 2,&1\le j\le8,\\ 3,&9\le j\le13,\\ 4,&14\le j\le15. \end{cases}$$

Third, it proves the Seed Echo Taxonomy Theorem, giving exact formulas for every clean echo set $\mathcal C_j$ and saturation threshold S_j for $j=0,\ldots,15$.

Fourth, it isolates the W9 Hinge Law: $W[9]$ is the first seed coordinate that opens the raw $t-7$ recurrence aperture, the first triple-entry seed word, and the length-sensitive hinge vertex of the final-block seed face $T_C=\{0,1,9\}$.

Fifth, it defines the projection layer through a fold-state vector $\Theta_9(r)$ and presents a testable transport hypothesis connecting recurrence clean echoes to attention nulls in transformer systems.

1.2 Security posture

The results in this paper are structural. They identify exact recurrence invariants in the SHA-256 message schedule. They do not demonstrate a break of SHA-256, a practical preimage recovery method, or a collision construction. The digest remains the output of a nonlinear 64-round compression process. The paper's claim is that part of the schedule's internal propagation geometry is exactly classifiable before downstream mixing saturates.

2. SHA-256 Schedule Preliminaries

SHA-256 operates on 512-bit message blocks. Each block is parsed into sixteen 32-bit words:

$$W[0],W[1],\ldots,W[15].$$

The schedule expands these to $W[0],\ldots,W[63]$ by setting the first sixteen words from the block and computing, for $16\le t\le63$,

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}.$$

The small sigma functions are:

$$\sigma_0(x)=\operatorname{ROTR}^{7}(x)\oplus\operatorname{ROTR}^{18}(x)\oplus\operatorname{SHR}^{3}(x),$$

$$\sigma_1(x)=\operatorname{ROTR}^{17}(x)\oplus\operatorname{ROTR}^{19}(x)\oplus\operatorname{SHR}^{10}(x).$$

This paper studies the dependency structure induced by the four lag positions:

$$t-2, \quad t-7, \quad t-15, \quad t-16.$$

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Only the schedule recurrence is analyzed. The compression function state variables (a,b,c,d,e,f,g,h) enter the later projection layer but are not required for the entry taxonomy.

2.1 Seed window and expansion zone

Define the seed window:

$$\mathcal S=\{0,1,\ldots,15\}.$$

Define the expansion zone:

$$\mathcal E=\{16,17,\ldots,63\}.$$

For any seed coordinate $j\in\mathcal S$, the question is:

> At which expansion coordinates does $W[j]$ appear directly, before being routed through another expanded word?

This question produces the direct-entry set.

3. The Aperture Complement Law

3.1 General form

Consider a recurrence of seed width N with lag set

$$L=\{\ell_1,\ldots,\ell_m\}.$$

The first expansion coordinate is N. A seed word at index j reaches the expansion zone through lag ℓ if

$$j+\ell\ge N.$$

Therefore the activation threshold for lag ℓ is

$$j\ge N-\ell.$$

This motivates the aperture complement.

Definition 3.1 (Aperture Complement). For a recurrence with seed width N and lag set L, define the aperture complement

$$A=N-L=\{N-\ell:\ell\in L\}.$$

The elements of A are the seed coordinates at which recurrence slots first become visible in the expansion zone.

3.2 SHA-256 aperture complement

For SHA-256,

$$N=16,\ \qquad L=\{2,7,15,16\}.$$

Therefore:

$$16-L=\{14,9,1,0\}.$$

In increasing seed order:

$$\boxed{A=\{0,1,9,14\}.}$$

The coordinates mean:

$$\begin{array}{c|c|c} \text{aperture coordinate} & \text{lag activated} & \text{slot}\\ \hline 0 & 16 & t-16\text{ raw slot}\\ 1 & 15 & t-15\text{ via }\sigma_0\\ 9 & 7 & t-7\text{ raw slot}\\ 14 & 2 & t-2\text{ via }\sigma_1 \end{array}$$

These four thresholds generate the partition

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$$\boxed{0\ \vert\ 1..8\ \vert\ 9..13\ \vert\ 14..15.}$$

This is the seed-window aperture map.

4. Direct-Entry Spectrum

4.1 Direct-entry set

Definition 4.1 (Direct-Entry Set). For a seed coordinate $j\in\{0,\ldots,15\}$, the direct-entry set $D(j)$ is the set of expansion indices $t\ge16$ at which $W[j]$ appears directly in the recurrence

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}.$$

Solving each slot for t yields four possible direct entries:

$$\begin{aligned} &t-16=j &&\Rightarrow &&t=j+16,\\ &t-15=j &&\Rightarrow &&t=j+15,\\ &t-7=j &&\Rightarrow &&t=j+7,\\ &t-2=j &&\Rightarrow &&t=j+2. \end{aligned}$$

The entry must satisfy $t\ge16$. Therefore:

$$\boxed{ D(j)=\{j+16\}\cup\{j+15:j\ge1\}\cup\{j+7:j\ge9\}\cup\{j+2:j\ge14\}. }$$

The corresponding transformations are:

$$\begin{array}{c|c|c} \text{entry} & \text{recurrence slot} & \text{form}\\ \hline j+16 & t-16 & W[j]\text{ raw}\\ j+15 & t-15 & \sigma_0(W[j])\\ j+7 & t-7 & W[j]\text{ raw}\\ j+2 & t-2 & \sigma_1(W[j]) \end{array}$$

4.2 Theorem: Direct-Entry Spectrum

Theorem 4.2 (Direct-Entry Spectrum). For the SHA-256 message schedule,

$$|D(j)|= \begin{cases} 1,&j=0,\\ 2,&1\le j\le 8,\\ 3,&9\le j\le 13,\\ 4,&14\le j\le 15. \end{cases}$$

Proof. The $t-16$ slot is always active because $j+16\ge16$ for every $j\ge0$. The $t-15$ slot is active exactly when $j+15\ge16$, i.e. $j\ge1$. The $t-7$ slot is active exactly when $j+7\ge16$, i.e. $j\ge9$. The $t-2$ slot is active exactly when $j+2\ge16$, i.e. $j\ge14$. Counting active slots gives the four cases. $\square$

4.3 Direct-entry table

$$\begin{array}{c|c|c|c} j & \text{class} & D(j) & |D(j)|\\ \hline 0 & \text{single} & \{16\} & 1\\ 1 & \text{double} & \{16,17\} & 2\\ 2 & \text{double} & \{17,18\} & 2\\ 3 & \text{double} & \{18,19\} & 2\\ 4 & \text{double} & \{19,20\} & 2\\ 5 & \text{double} & \{20,21\} & 2\\ 6 & \text{double} & \{21,22\} & 2\\ 7 & \text{double} & \{22,23\} & 2\\ 8 & \text{double} & \{23,24\} & 2\\ 9 & \text{triple} & \{16,24,25\} & 3\\ 10 & \text{triple} & \{17,25,26\} & 3\\ 11 & \text{triple} & \{18,26,27\} & 3\\ 12 & \text{triple} & \{19,27,28\} & 3\\ 13 & \text{triple} & \{20,28,29\} & 3\\ 14 & \text{quad} & \{16,21,29,30\} & 4\\ 15 & \text{quad} & \{17,22,30,31\} & 4 \end{array}$$

The coordinate $j=9$ is the first triple-entry coordinate because it is the first coordinate at which the raw $t-7$ slot reaches the expansion zone:

$$9+7=16.$$

5. Contamination Recurrence and Clean Echo Sets

5.1 Contamination state

The direct-entry spectrum describes where a seed word first enters the expansion zone. We now ask which later expanded words become dependent on a chosen seed perturbation.

Fix $j\in\{0,\ldots,15\}$. Define a binary contamination state $C_t^{(j)}$ by:

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$$C_t^{(j)}= \begin{cases} 1,&W[t]\text{ depends on }W[j],\\ 0,&W[t]\text{ is clean of }W[j]. \end{cases}$$

The seed condition is:

$$C_j^{(j)}=1, \qquad C_k^{(j)}=0\quad(k\ne j,\,0\le k\le 15).$$

For $t\ge16$, the schedule recurrence induces the logical contamination recurrence:

$$\boxed{ C_t^{(j)}=C_{t-2}^{(j)}\lor C_{t-7}^{(j)}\lor C_{t-15}^{(j)}\lor C_{t-16}^{(j)}. }$$

The transformations σ_0 and σ_1 are irrelevant to this binary dependency recurrence because they do not erase dependency.

5.2 Clean echo set

Definition 5.1 (Clean Echo Set). The clean echo set for seed word $W[j]$ is

$$\mathcal C_j=\{t\in\{16,\ldots,63\}:C_t^{(j)}=0\}.$$

It is the negative image of the contamination cone: the expanded coordinates that a perturbation at $W[j]$ cannot reach.

5.3 Saturation threshold

Definition 5.2 (Saturation Threshold). The saturation threshold S_j is the first expanded index s such that

$$C_s^{(j)}=1 \qquad\text{and}\qquad C_{s+1}^{(j)}=1.$$

The pair (S_j,S_j+1) is called the first consecutive dirty pair.

5.4 Saturation lemma

Lemma 5.3 (Consecutive Dirty Pair Saturation). If $C_s^{(j)}=1$ and $C_{s+1}^{(j)}=1$ for some $s\ge16$, then

$$C_t^{(j)}=1\qquad\text{for all }t\ge s.$$

up to the convention that s and $s+1$ start the saturated interval.

Proof. Since $C_s^{(j)}=1$, the $t-2$ input contaminates $C_{s+2}^{(j)}$. Since $C_{s+1}^{(j)}=1$, the $t-2$ input contaminates $C_{s+3}^{(j)}$. Inductively, if $C_u^{(j)}=1$ and $C_{u+1}^{(j)}=1$, then the $t-2$ slot contaminates $C_{u+2}^{(j)}$ and $C_{u+3}^{(j)}$. Therefore every later index is contaminated. $\square$

This lemma explains why the first consecutive dirty pair is the decisive event. A single dirty ray is not yet a sheet. Two adjacent dirty coordinates generate a sheet through the $t-2$ recurrence slot.

6. The Seed Echo Taxonomy Theorem

This section classifies $\mathcal C_j$ and S_j for every seed coordinate.

6.1 Case $j=0$: single-entry origin

For $j=0$,

$$D(0)=\{16\}.$$

The first dirty coordinate is 16. The $t-2$ arm gives

$$16\to18\to20\to22.$$

The $t-7$ arm from 16 arrives at

$$16+7=23.$$

Therefore the first consecutive dirty pair is

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$$(22,23),$$

and

$$S_0=22.$$

The clean words before saturation are the odd interstices:

$$\boxed{\mathcal C_0=\{17,19,21\}.}$$

6.2 Case $1\le j\le8$: double-entry class

For $1\le j\le8$,

$$D(j)=\{j+15,j+16\}.$$

These two direct entries are consecutive. Therefore the direct-entry pair itself is already the first consecutive dirty pair:

$$S_j=j+15.$$

Every expansion coordinate before $j+15$ is clean. Hence:

$$\boxed{\mathcal C_j=\{16,\ldots,j+14\},\qquad 1\le j\le8.}$$

The cardinality is

$$|\mathcal C_j|=j-1.$$

The edge case $j=1$ gives

$$D(1)=\{16,17\},\quad \mathcal C_1=\emptyset,\quad S_1=16.$$

Thus $W[1]$ is the unique seed word with zero clean echo words.

6.3 Case $9\le j\le13$: triple-entry class

For $9\le j\le13$,

$$D(j)=\{j+7,j+15,j+16\}.$$

Let

$$a_j=j+7$$

be the first direct dirty point. The fast $t-2$ arm contaminates

$$a_j,\quad a_j+2,\quad a_j+4,\quad a_j+6,\ldots$$

or equivalently

$$j+7,\quad j+9,\quad j+11,\quad j+13,\ldots$$

The slow $t-7$ arm from a_j reaches

$$a_j+7=j+14.$$

Therefore the first consecutive dirty pair is

$$(j+13,j+14),$$

and

$$S_j=j+13.$$

The clean echo set consists of the pre-saturation expansion coordinates with the fast-arm dirty points removed:

$$\boxed{\mathcal C_j=\{16,\ldots,j+12\}\setminus\{j+7,j+9,j+11\},\qquad 9\le j\le13.}$$

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The cardinality is

$$|\mathcal C_j|=j-6.$$

The last clean word in this class is $j+12$, while the third direct entry is $j+16$. Therefore the boundary gap is universal:

$$(j+16)-(j+12)=4.$$

6.4 Case $j=14,15$: quad-entry class

For $j=14$,

$$D(14)=\{16,21,29,30\}.$$

The $t-2$ slot activates immediately at 16. The fast arm gives 18 and 20. The raw $t-7$ direct entry fires at 21. Thus the first consecutive dirty pair is

$$(20,21),$$

so

$$S_{14}=20, \qquad \boxed{\mathcal C_{14}=\{17,19\}.}$$

For $j=15$,

$$D(15)=\{17,22,30,31\}.$$

The $t-2$ slot activates at 17. The fast arm gives 19 and 21. The raw $t-7$ entry fires at 22. Thus the first consecutive dirty pair is

$$(21,22),$$

so

$$S_{15}=21, \qquad \boxed{\mathcal C_{15}=\{16,18,20\}.}$$

6.5 Theorem statement

Theorem 6.1 (Seed Echo Taxonomy). Under the SHA-256 message schedule recurrence, the clean echo set $\mathcal C_j$ and saturation threshold S_j are exactly:

$$\begin{array}{c|c|c} \text{case} & \mathcal C_j & S_j\\ \hline j=0 & \{17,19,21\} & 22\\ 1\le j\le8 & \{16,\ldots,j+14\} & j+15\\ 9\le j\le13 & \{16,\ldots,j+12\}\setminus\{j+7,j+9,j+11\} & j+13\\ j=14 & \{17,19\} & 20\\ j=15 & \{16,18,20\} & 21 \end{array}$$

Proof. The five cases are derived in Sections 6.1-6.4 by applying the contamination recurrence and identifying the first consecutive dirty pair. Lemma 5.3 then forces all later coordinates dirty. $\square$

7. Unified Saturation Matrix

The full classification is shown below.

$$\begin{array}{c|c|c|c|c} j & \text{class} & D(j) & \mathcal C_j & |\mathcal C_j| & S_j\\ \hline 0 & \text{single} & \{16\} & \{17,19,21\} & 3 & 22\\ 1 & \text{double} & \{16,17\} & \emptyset & 0 & 16\\ 2 & \text{double} & \{17,18\} & \{16\} & 1 & 17\\ 3 & \text{double} & \{18,19\} & \{16,17\} & 2 & 18\\ 4 & \text{double} & \{19,20\} & \{16,17,18\} & 3 & 19\\ 5 & \text{double} & \{20,21\} & \{16,17,18,19\} & 4 & 20\\ 6 & \text{double} & \{21,22\} & \{16,17,18,19,20\} & 5 & 21\\ 7 & \text{double} & \{22,23\} & \{16,17,18,19,20,21\} & 6 & 22\\ 8 & \text{double} & \{23,24\} & \{16,17,18,19,20,21,22\} & 7 & 23\\ 9 & \text{triple} & \{16,24,25\} & \{17,19,21\} & 3 & 22\\ 10 & \text{triple} & \{17,25,26\} & \{16,18,20,22\} & 4 & 23\\ 11 & \text{triple} & \{18,26,27\} & \{16,17,19,21,23\} & 5 & 24\\ 12 & \text{triple} & \{19,27,28\} & \{16,17,18,20,22,24\} & 6 &$$

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$$25\\ 13 & \text{triple} & \{20,28,29\} & \{16,17,18,19,21,23,25\} & 7 & 26\\ 14 & \text{quad} & \{16,21,29,30\} & \{17,19\} & 2 & 20\\ 15 & \text{quad} & \{17,22,30,31\} & \{16,18,20\} & 3 & 21 \end{array}$$

7.1 Saturation equivalence class at $S=22$

A striking feature of the taxonomy is

$$\boxed{S_0=S_7=S_9=22.}$$

The equality hides three distinct mechanisms.

For $j=0$, the schedule saturates through arm timing from a single entry at 16:

$$16\to18\to20\to22$$

and

$$16\to23.$$

For $j=7$, the schedule saturates immediately because

$$D(7)=\{22,23\}.$$

For $j=9$, the schedule saturates through the collision of the fast and slow arms from the first triple-entry point 16:

$$16\to18\to20\to22$$

and

$$16\to23.$$

Thus $S=22$ is a route-equivalence class: distinct seed-entry mechanisms collapse onto the same saturation address.

8. The W9 Hinge Law

8.1 Why $W[9]$ is structurally distinguished

The coordinate $W[9]$ is distinguished by two properties.

First, it is the first triple-entry seed word:

$$D(9)=\{16,24,25\}.$$

Second, it is the length-sensitive hinge vertex of the final-block seed triangle

$$T_C=\{0,1,9\}.$$

The clean echo pattern $\{17,19,21\}$ is not unique to $W[9]$ because $W[0]$ shares it. The uniqueness of $W[9]$ is the conjunction:

$$\boxed{ W[9]=\text{first triple-entry seed word} + \text{length-sensitive hinge vertex}. }$$

8.2 Three direct entries of $W[9]$

The three entries are:

$$W[16]=\sigma_1(W[14])+W[9]+\sigma_0(W[1])+W[0],$$
$$W[24]=\sigma_1(W[22])+W[17]+\sigma_0(W[9])+W[8],$$
$$W[25]=\sigma_1(W[23])+W[18]+\sigma_0(W[10])+W[9].$$

Thus $W[9]$ strikes the expansion lattice as:

$$\boxed{\{\text{raw at }16,\ \sigma_0\text{ at }24,\ \text{raw at }25\}.}$$

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8.3 Triple convergence at $W[25]$

At $W[25]$:

$$W[25]=\sigma_1(W[23])+W[18]+\sigma_0(W[10])+W[9].$$

Here:

- $W[18]$ carries the fast-arm contamination from $W[16]$.
- $W[23]$ carries the slow-arm contamination from $W[16]$.
- $W[9]$ re-enters raw through the $t-16$ slot.
- $W[10]$ is the clean reference input.

Therefore:

$$\boxed{ W[25]=\text{fast arm}+\text{slow arm}+\text{raw re-entry}+\text{clean reference}. }$$

The envelope $16\to25$ has span

$$25-16=9.$$

The 8-word window $W[16..23]$ is the interior of this 9-step hinge envelope. $W[25]$ is the closure point outside the window.

9. First Aperture Readout and Final-Block Seed Crystal

The first expanded word is

$$W[16]=\sigma_1(W[14])+W[9]+\sigma_0(W[1])+W[0].$$

This is the first full aperture readout of the seed skeleton

$$\boxed{A_{16}=\{0,1,9,14\}.}$$

For the common final-block case in which $W[14]=0$, this collapses to

$$\boxed{W[16]=W[0]+\sigma_0(W[1])+W[9]\pmod{2^{32}}.}$$

Thus the active content face is

$$\boxed{T_C=\{0,1,9\}.}$$

The gap signature of this face is

$$0\to1=1,\ \qquad 1\to9=8,\ \qquad 9\to16=7,$$

so

$$\boxed{T_C=(1,8,7),\qquad 1+8+7=16.}$$

The observer or length face is

$$\boxed{T_O=\{1,9,15\}.}$$

Its gaps are

$$1\to9=8,\ \qquad 9\to15=6,\ \qquad 15\to17=2,$$

so

$$\boxed{T_O=(8,6,2),\qquad 8+6+2=16.}$$

The shared hinge is

$$\boxed{1\to9,\ \qquad 9-1=8=16/2.}$$

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The two-triangle picture is therefore a projection of the four-point aperture $\{0,1,9,14\}$ after the $W[14]$ seam collapses.

9.1 Scope condition on $W[14]=0$

The identity

$$W[16]=W[0]+\sigma_0(W[1])+W[9]$$

requires

$$W[14]=0.$$

In standard SHA-256 padding, $W[14]$ is the high 32-bit word of the 64-bit message length in the final block. Therefore $W[14]=0$ for messages whose bit length is less than 2^{32}. For arbitrary non-final blocks, or for final blocks of very long messages where the high length word is nonzero, the general four-point aperture must be retained:

$$W[16]=\sigma_1(W[14])+W[9]+\sigma_0(W[1])+W[0].$$

This scope condition is essential. The three-point face $T_C=\{0,1,9\}$ is the collapsed final-block face, not the universal arbitrary-block aperture.

10. The $W[16]$ Divergence Law

For two schedule instances a and b satisfying

$$W_a[0]=W_b[0], \qquad W_a[1]=W_b[1], \qquad W_a[14]=W_b[14]=0,$$

we have:

$$W_a[16]-W_b[16] \equiv W_a[9]-W_b[9] \pmod{2^{32}}.$$

Equivalently,

$$\boxed{ \Delta_{\mathbb Z/2^{32}}W[16]=\Delta_{\mathbb Z/2^{32}}W[9]. }$$

This law is not a statement about the final digest Hamming distance. It is a pre-avalanche schedule diagnostic. The final digest may show ordinary avalanche, but the first aperture readout preserves the signed modular differential exactly under the stated conditions.

In the general case where $W[14]$ is not fixed or not zero,

$$\Delta W[16] \equiv \Delta\sigma_1(W[14])+ \Delta W[9]+ \Delta\sigma_0(W[1])+ \Delta W[0] \pmod{2^{32}}.$$

The $W[9]$ divergence law is therefore a controlled-isolation law, not an unrestricted arbitrary-message identity.

11. Projection Machine and Fold-State Vector

The aperture and echo machines are recurrence-internal. The projection machine connects message format to schedule face selection.

Let r denote the number of message content bytes carried into the final block before the padding sentinel $0x80$ is appended, with

$$0\le r\le 55$$

for a final block that contains padding and length in the same block.

The word $W[9]$ occupies byte positions

$$36,37,38,39.$$

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Define:

$$c_9(r)=\min(4,\max(0,r-36)),$$

$$p_9(r)= \begin{cases} 1,&36\le r\le39,\\ 0,&\text{otherwise}, \end{cases}$$

and

$$z_9(r)=4-c_9(r)-p_9(r).$$

The fold-state vector is

$$\boxed{ \Theta_9(r)=\frac{1}{4}(c_9(r),p_9(r),z_9(r)). }$$

This gives three discrete face states:

$$\begin{array}{c|c|c} \text{range} & W[9]\text{ state} & \text{face}\ \hline r<36 & 0x00000000 & \text{zero face}\\ 36\le r\le39 & \text{contains }0x80\text{ sentinel} & \text{padding crease}\\ r\ge40 & \text{fully occupied by content} & \text{live content face} \end{array}$$

The threshold $r=40$ is the point at which $W[9]$ becomes fully live. This is also the byte depth at which all three vertices of the collapsed content face

$$T_C=\{0,1,9\}$$

can be fully occupied by message content.

11.1 On projection angles

The recurrence theorems above do not require any geometric angle assignment. If angles such as 0°, a padding crease angle, or a proposed live-content angle such as 156° are used, they must be defined by an explicit projection functional

$$\alpha(r)=\mathcal P(\Theta_9(r),T_C,T_O)$$

or measured empirically from trace vectors. Until such a projection functional is specified, the angle layer should be treated as an empirical projection hypothesis, not as part of the proven recurrence taxonomy.

A suitable empirical trace vector is:

$$P_9(r)= \big(HD(W[16]), HD(W[24]), HD(W[25]), HD(S_8), HD(S_{16}), HD(S_{25}), HD(h) \big),$$

where S_t denotes the compression state after round t and h denotes the final digest. This vector preserves intermediate geometry before final avalanche saturates the measurement.

12. Three Nested Machines

The entire structure can now be compressed as three nested machines.

12.1 Aperture Machine

The aperture machine is determined by

$$L=\{2,7,15,16\}$$

and

$$A=16-L=\{0,1,9,14\}.$$

It defines the direct-entry classes and the first aperture readout at $W[16]$.

12.2 Echo and Saturation Machine

The echo/saturation machine is governed by

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$$C_t^{(j)}=C_{t-2}^{(j)}\lor C_{t-7}^{(j)}\lor C_{t-15}^{(j)}\lor C_{t-16}^{(j)}.$$

It produces clean echo sets $\mathcal C_j$ and saturation thresholds S_j.

12.3 Projection and Shadow Machine

The projection machine maps byte-level final-block structure into seed-face states through

$$\Theta_9(r)=\frac{1}{4}(c_9(r),p_9(r),z_9(r)).$$

The final digest is a shadow of this selected face after compression-round mixing. In symbolic form:

$$\boxed{ \text{lag set}\to\text{aperture complement}\to\text{seed face}\to\text{clean echo}\to\text{saturation sheet}\to\text{digest projection}. }$$

13. Transformer Attention Transport Hypothesis

This section is a transport hypothesis, not a theorem. It maps the SHA-256 recurrence geometry into an analogous structure in transformer attention. The purpose is to define testable correspondences, not to assert that SHA-256 and transformers are identical systems.

13.1 Structural correspondence

$$\begin{array}{c|c|c} \text{SHA-256 structure} & \text{Transformer analogue} & \text{functional role}\\ \hline W[0],W[1] & W_K,W_V\text{ or static projection weights} & \text{base manifold}\\ W[9] & Q\text{ or prompt-sensitive query direction} & \text{dynamic hinge}\\ W[15] & \text{position/mask/context boundary} & \text{observer boundary}\\ W[16] & \text{first visible token-state projection} & \text{first aperture readout}\\ \{17,19,21\} & \text{attention nulls or clean latent slots} & \text{negative geometry}\\ S_j & \text{context-global contamination threshold} & \text{saturation address}\\ \Theta_9(r) & \text{discrete attention head or routing state} & \text{face selection} \end{array}$$

13.2 Testable prediction

The transport hypothesis predicts that prompt perturbations in transformer systems should admit analogues of:

1. direct visibility events,
2. clean echo positions,
3. saturation thresholds,
4. route-equivalence classes,
5. projection face states.

A transformer experiment would perturb a selected prompt token or slot, trace attention and residual-stream changes layer by layer, and define a binary contamination recurrence over latent positions or channels. The model would then test whether certain positions remain clean before a saturation threshold, analogous to $\mathcal C_j$ in the SHA-256 schedule.

The current paper proves the SHA-256 side. The transformer side remains an empirical program.

14. Experimental Reproducibility Protocol

The recurrence taxonomy can be verified without implementing the full SHA-256 compression function. The following pseudocode suffices.

```
def direct_entries(j):
    out = {j + 16}
    if j >= 1:
        out.add(j + 15)
```

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```
        if j >= 9:
            out.add(j + 7)
        if j >= 14:
            out.add(j + 2)
        return sorted(out)


    def contamination_trace(j, n=64):
        C = [0] * n
        C[j] = 1
        for t in range(16, n):
            C[t] = C[t-2] or C[t-7] or C[t-15] or C[t-16]
        clean = [t for t in range(16, n) if C[t] == 0]
        saturation = None
        for t in range(16, n-1):
            if C[t] and C[t+1]:
                saturation = t
                break
        return clean, saturation, C
```

A valid verification run checks:

$$D(j),\quad \mathcal C_j,\quad S_j$$

for every

$$j=0,1,\ldots,15.$$

Expected results are exactly the entries in the unified saturation matrix in Section 7.

15. Limitations and Open Problems

15.1 Arbitrary-block aperture

The collapsed face $T_C=\{0,1,9\}$ assumes $W[14]=0$. For arbitrary non-final blocks or final blocks with nonzero high length word, the full aperture

$$\{0,1,9,14\}$$

must be retained.

15.2 Projection angle derivation

The angle language is useful geometrically but must be tied to an explicit projection functional or empirical trace measurement. A proposed 156° live-content angle should be treated as a hypothesis until derived from $\Theta_9(r)$ or from trace-vector geometry.

15.3 Differential cryptanalysis relevance

The clean echo sets show exact locations that a single seed perturbation does not reach before saturation. Future work should determine whether these negative-geometry zones can improve differential schedule analysis or merely describe dependency timing without practical security consequence.

15.4 Clean echo intersection lattice

The sets $\mathcal C_j$ have nontrivial intersections. For example:

$$\mathcal C_0\cap\mathcal C_9=\{17,19,21\}.$$

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A complete intersection lattice may reveal higher-order echo nodes in the schedule.

15.5 Transformer-side verification

The transformer transport hypothesis requires separate empirical testing. The existence of a formal SHA-256 recurrence taxonomy does not by itself prove that transformer attention has the same clean echo structure.

16. Conclusion

Aperture Threshold Geometry reframes the SHA-256 message schedule as a deterministic recurrence machine with exact visibility thresholds and saturation boundaries. The lag set

$$L=\{2,7,15,16\}$$

reflects against the 16-word seed width to produce the aperture complement

$$A=\{0,1,9,14\}.$$

This complement partitions the seed window into single-, double-, triple-, and quad-entry classes. From this partition the Direct-Entry Spectrum Theorem follows immediately. The binary contamination recurrence then yields the Seed Echo Taxonomy Theorem, classifying every clean echo set and saturation threshold for $j=0,\ldots,15$.

The coordinate $W[9]$ is a distinguished hinge because it is the first triple-entry seed word and the length-sensitive vertex of the final-block content face $T_C=\{0,1,9\}$. Its three direct entries at $16,24,25$ create a 9-step hinge envelope whose interior is the Sziklai window $W[16..23]$ and whose closure is $W[25]$.

The essential map is:

$$\boxed{ \text{lag set} \to \text{aperture complement} \to \text{entry class} \to \text{clean echo} \to \text{saturation threshold} \to \text{projection shadow}. }$$

This is the top-down view. SHA-256 expansion is not merely diffusion. It is aperture selection followed by echo formation, saturation, and projection.

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Appendix A. Full Clean Echo Taxonomy in Compact Form

$$\begin{aligned} \mathcal C_0&=\{17,19,21\},&S_0&=22,\\ \mathcal C_j&=\{16,\ldots,j+14\},&S_j&=j+15,\quad 1\le j\le8,\\ \mathcal C_j&=\{16,\ldots,j+12\}\setminus\{j+7,j+9,j+11\},&S_j&=j+13,\quad 9\le j\le13,\\ \mathcal C_{14}&=\{17,19\},&S_{14}&=20,\\ \mathcal C_{15}&=\{16,18,20\},&S_{15}&=21. \end{aligned}$$

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Appendix B. Exact $W[9]$ Entry Equations

$$W[16]=\sigma_1(W[14])+W[9]+\sigma_0(W[1])+W[0].$$

$$W[24]=\sigma_1(W[22])+W[17]+\sigma_0(W[9])+W[8].$$

$$W[25]=\sigma_1(W[23])+W[18]+\sigma_0(W[10])+W[9].$$

For $W[14]=0$:

$$W[16]=W[0]+\sigma_0(W[1])+W[9].$$

If $W[0]$ and $W[1]$ are fixed:

$$\Delta W[16]\equiv\Delta W[9]\pmod{2^{32}}.$$

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